

Simulation Results

Smart-Root–Dumb-Tree Model

Search scope : Combat move selection only - attacks only 1 territory in a round

Selection policy : UCB1 with exploration constant $c = 1.414$

Expansion : Single node per iteration

Simulation policy : Uniform random rollout for all players - keeps the tree dumb and root smart - excludes purchase and placement delegates - the simulation move generation has no pruning apart from time budget limit

Forward model : Handles the logs from the game, provides the requested move for the MCTS player (tree selected for combat and random for other delegates)

State evaluation : Territory strength ration + depth based penalties for non-terminal states

Time budget : 1s per combat turn

Max tree depth : 10

Goal : Isolate and measure combat decision quality independent of the other game delegates

Legal Combat Move Generation

- Territories are ordered by priority before move generation, which implicitly biases the search toward stronger actions. Priority is determined using the victory condition, factory presence, and territorial vicinity.
- Search is performed on a per-territory basis, enumerating all feasible attack combinations for a given target territory.
- Attacks that are strategically disadvantageous are pruned by comparing aggregate attacking strength against the defending territory's strength. If a territory cannot be successfully conquered, it is excluded from further consideration.
- For undefended enemy territories, only a single attacking unit is selected (at random), as this is sufficient to capture the territory.
- The reachability of the units is cached, and cache is renewed every round.
- The move-generation function respects a predefined time budget and returns the best partial set of actions once the limit is reached. This ensures smooth gameplay execution while preserving the overall quality of MCTS decisions, which is evaluated through node quality metrics.

Legal Non-Combat Move Generation

- Non-combat movement is executed to further reposition units strategically across the game board.
- The move-generation function respects a predefined time budget and returns the best partial set of actions once the limit is reached. This ensures smooth gameplay execution while preserving

the overall quality of MCTS decisions, which is evaluated through node quality metrics.

- The reachability of the units is cached, and cache is renewed every round.

Efficiency summary

MCTS Efficiency Summary (Game-wise)

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Game 1

Total rounds: 29

Iterations per round:

Mean: 2.52

Min : 1

Max : 15

Exploration_coverage (actions_explored / total_actions):

Mean: 0.3124

Min : 0.0001

Max : 1.0000

Avg Tree depth per round:

Mean: 4.9000

Min : 2.0000

Max : 10.0000

Game 2

Total rounds: 22

Iterations per round:

Mean: 3.73

Min : 1

Max : 20

Exploration_coverage (actions_explored / total_actions):

Mean: 0.2924

Min : 0.0000

Max : 1.0000

Avg Tree depth per round:

Mean: 5.9765

Min : 1.0000

Max : 10.0000

Game 3

Total rounds: 45

Iterations per round:

Mean: 1.89

Min : 1

Max : 12

Exploration_coverage (actions_explored / total_actions):

Mean: 0.1534

Min : 0.0000

Max : 1.0000

Avg Tree depth per round:

Mean: 3.3248

Min : 1.0000

Max : 9.7500

Game 4

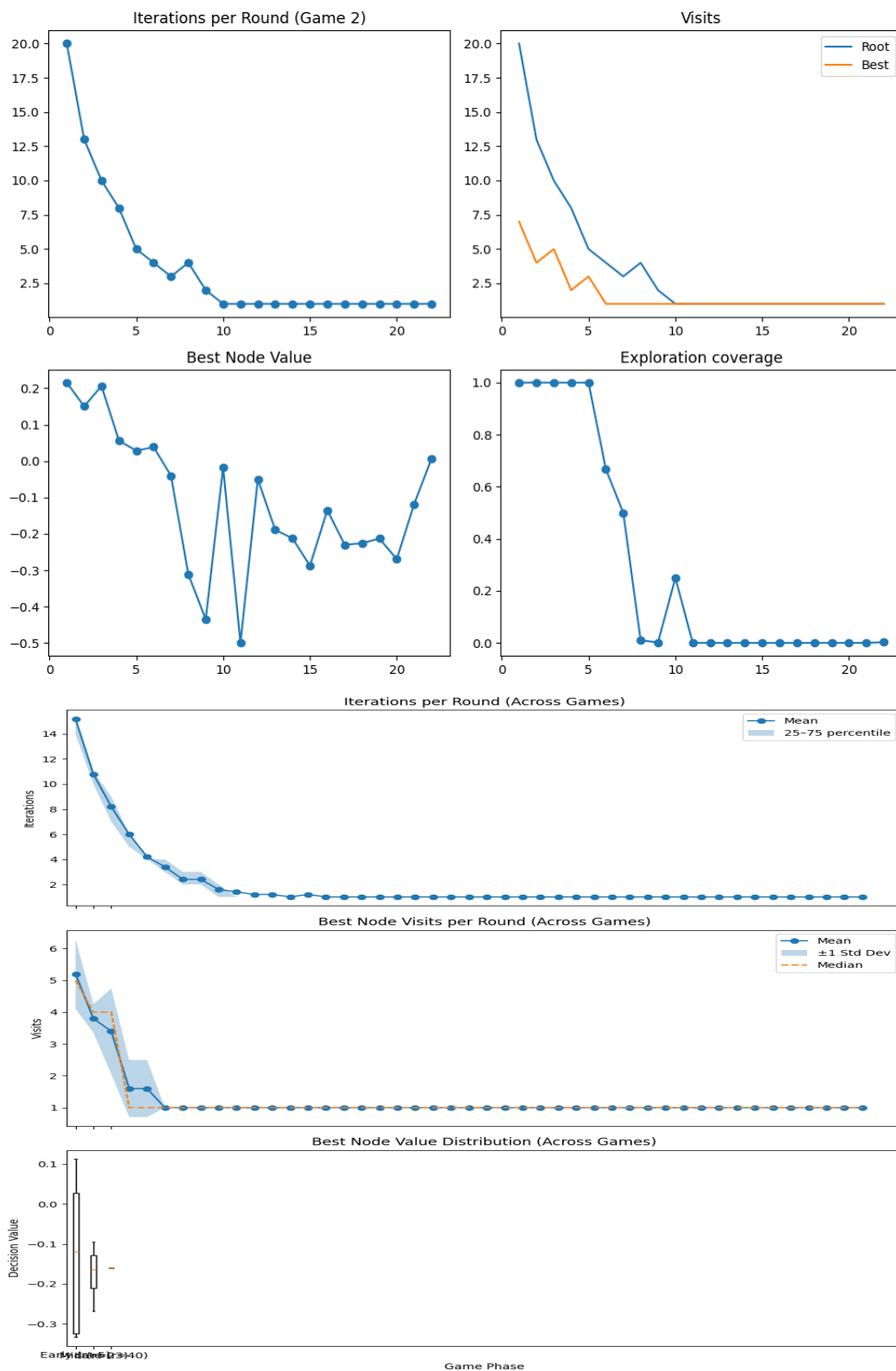
Total rounds: 35
Iterations per round:
Mean: 1.89
Min : 1
Max : 14
Exploration_coverage (actions_explored / total_actions):
Mean: 0.1492
Min : 0.0000
Max : 1.0000
Avg Tree depth per round:
Mean: 3.0670
Min : 1.0000
Max : 9.9286
Game 5

Total rounds: 29
Iterations per round:
Mean: 2.66
Min : 1
Max : 15
Exploration_coverage (actions_explored / total_actions):
Mean: 0.2293
Min : 0.0000
Max : 1.0000
Avg Tree depth per round:
Mean: 4.6651
Min : 1.0000
Max : 10.0000

OVERALL AVERAGES (Across Games)

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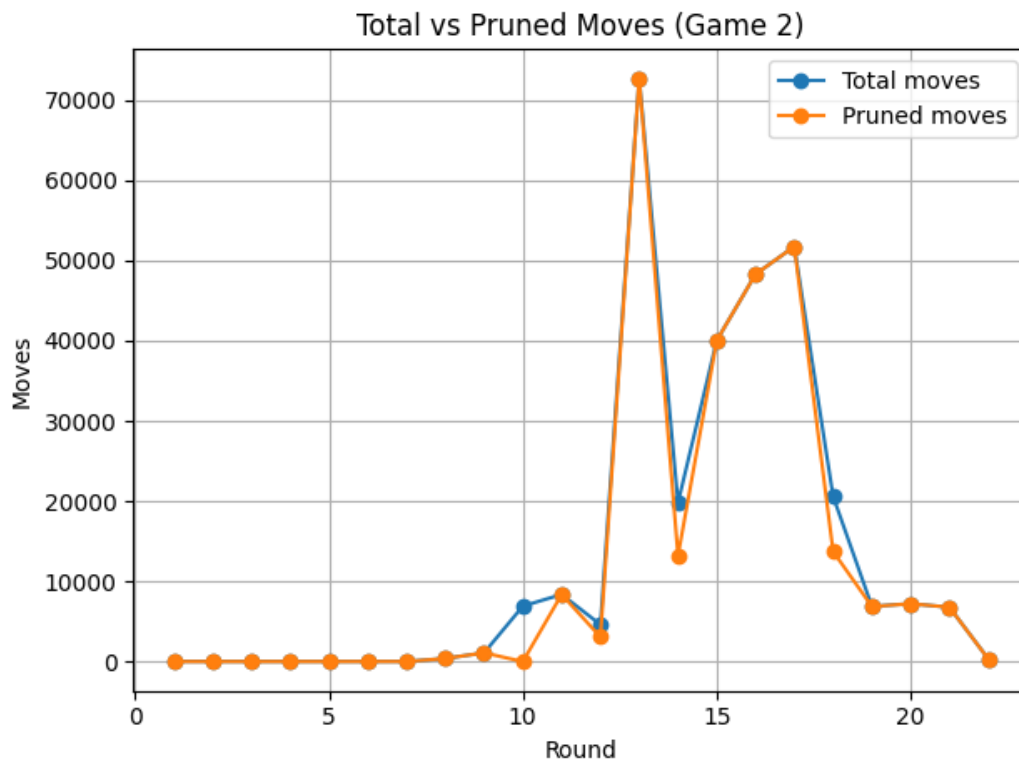
Iterations per round (game-averaged):
Mean: 2.53
Min: 1
Max: 20
Exploration coverage (game-averaged):
Mean: 0.23
Min: 0.00
Max: 1.0
Avg tree depth per round (game-averaged):
Mean: 4.39
Min: 1.0
Max: 10.0

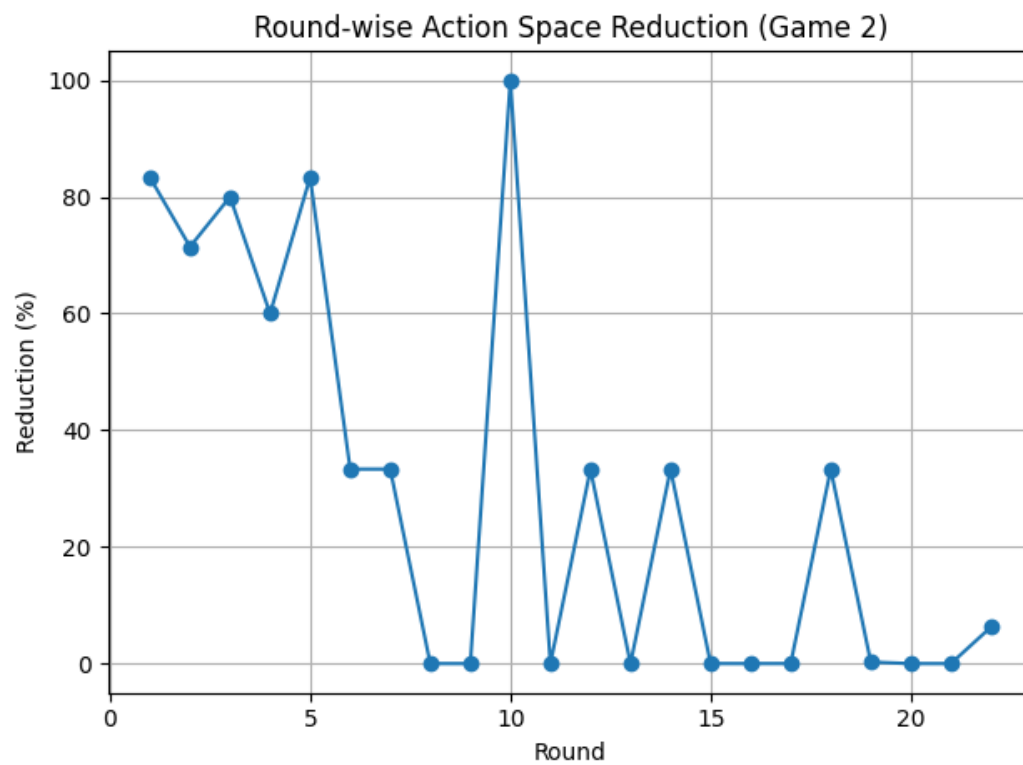


Action space reduction summary

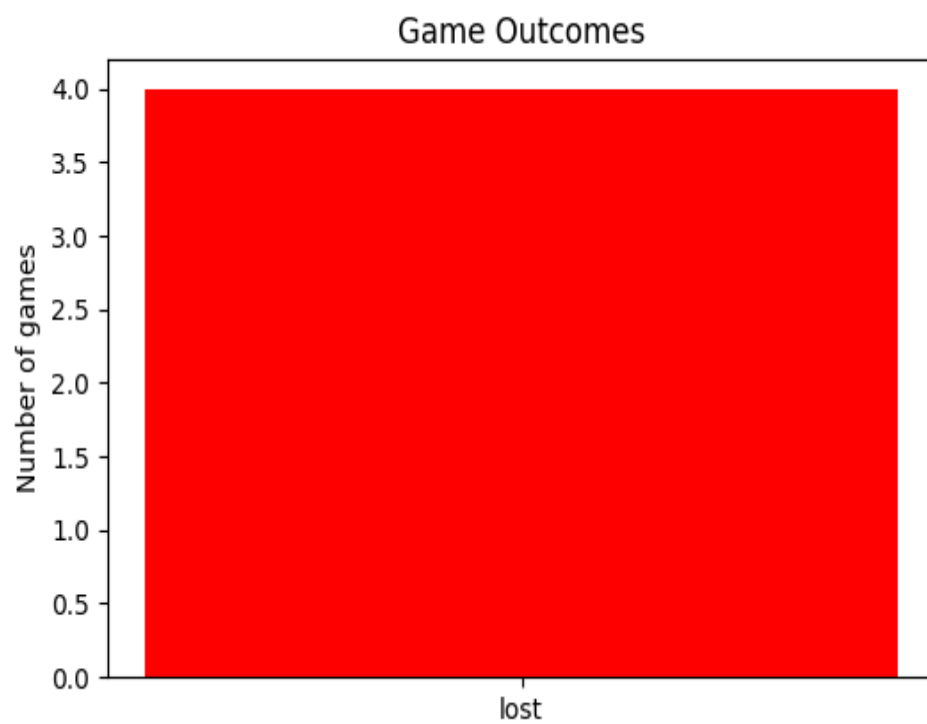
game	rounds	avg_round_reduction	avg_round_reduction_percent
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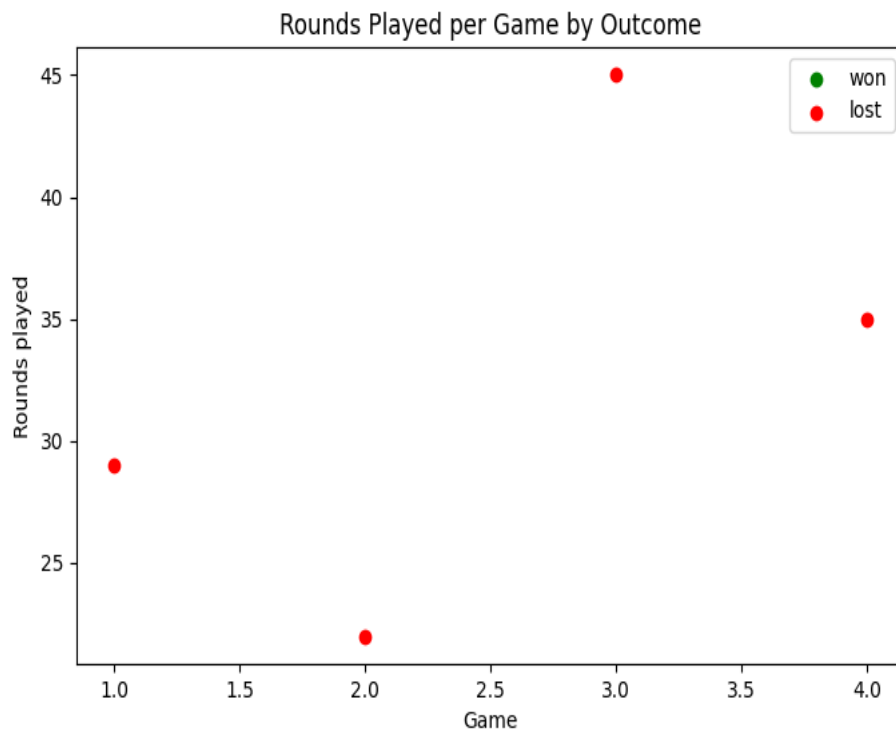
1	29.0	0.285104	28.510381
2	22.0	0.295981	29.598119
3	45.0	0.189538	18.953844
4	44.0	0.188566	18.856586
5	30.0	0.248214	24.821441
OVERALL	34.0	0.241481	24.148074





Game outcomes





Combat move quality throughout the games

=== Combat Quality Metric Averages ===

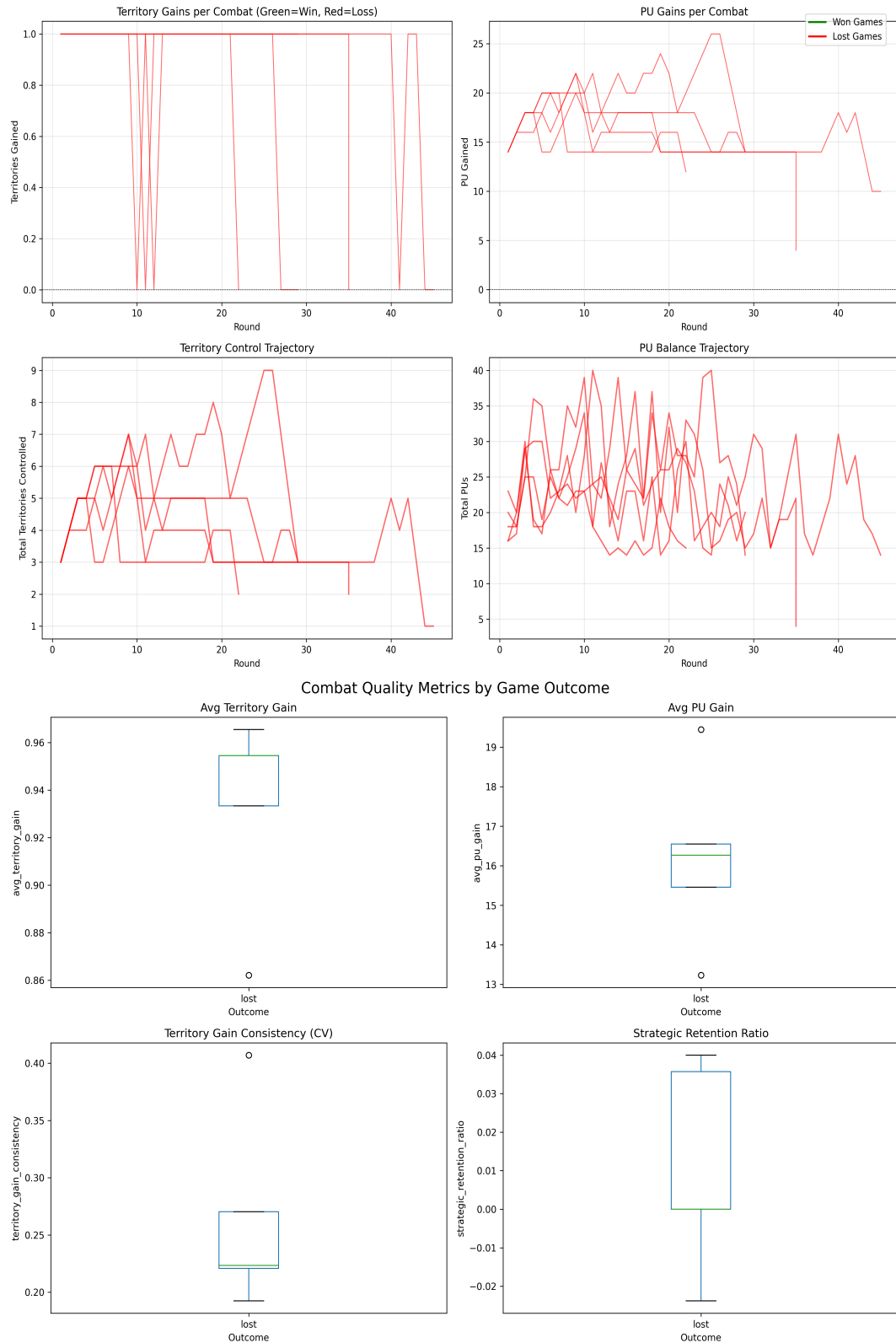
Outcome: LOST

Avg Territory Gain: 0.934

Avg PU Gain: 16.190

Territory Gain Consistency: 0.263

Strategic Retention Ratio: 0.010



Profiling of MCTS algorithm

